RAFFLES INSTITUTION
H2 Computing (9569)

2025 Year 5

Tutorial S3A: Data Representation (Solutions)

1. Characters are stored inside a computer as a **series of 0s and 1s**. A single 0 or 1 is called a binary digit or *bit.* Any group of 0s and 1s can be used to represent a specific character, **which can be a letter, number, or symbol**. Eight bits (or called a **byte**) are used to store one character. **For example**, the character 'A' can be represented by these eight bits as 01000001, which is 41(hex). The complete set of characters that the computer uses is known as its **character set**.

2. Differences between ASCII and Unicode

ASCII	Unicode
7-bit character set, total 128 characters. Limited set includes uppercase & lowercase letters, numbers, punctuation marks, and control characters.	Unicode supports up to 4 bytes character set, allow for the representation of thousands of characters from various scripts and languages.
Fixed-length encoding scheme – 7 bit binary value	Variable-length encoding schemes, such as UTF-8, UTF-16, and UTF-32. Unicode-encoded text files may be larger in size compared to ASCII-encoded files.
Primarily supports English and a few other Western languages.	Provides extensive support for multiple languages, including non-Latin scripts like Cyrillic, Arabic, Chinese, Japanese, and many more. More inclusive and versatile standard for multilingual text representation. Localisation of software.

(Ensure working is shown in all conversion of different number bases)

3. (a) 1010 (10) (b) 101000 (40) (c) 11111111 (255) (d) 010100111011 **(1339)**

4. (a) 27 (11011) (b) 128 (10000000) (c) 789 (1100010101)
   (d) 1768 **(11011101000)** [show long division method]

5. (a)
```
  1011
 +1101
 11000
```
In denary: 2^4+2^3 = 24

(b)
```
  100101
 +101011
 1010000
```
In denary: 2^6+2^4 = 64+16 = 80

6. (a) 10101011 (AB) (b) 11110000 (F0)
   (c) 1001100101010100 (9954) (d) 1010101 (55)

7. (a) 160 (101100000) (b) 279 (100010111)
   (c) EF60 (1110111101100000) (d) ABC3 (1010101111000011)

8. Denary 1567 = Hex 61F

16 |1567
16 |__97 rem 15 (F)
16 |___6 rem 1
0 rem 6

9. Hex 2A9D = $2 \times 16^3 + 10 \times 16^2 + 9 \times 16^1 + 13 \times 16^0$ = 10909 in denary

10.(a) 'D' in binary as a byte: 01000100 (b) 'D' in hex: 44

11. Hex code is 4E2D. Binary is 0100 1110 0010 1101.

12. Red = hex 4C = 4x16 + 12 = 64 + 12 = 76
    Green = hex 99 = 9x16 + 9 = 144 + 9 = 153
    Blue = hex FF = 15x16 + 15 = 240 + 15 = 255

13. (a) Hex B7C3 = 1011 0111 1100 0011 in binary
    (b) Long binary number is harder to read. Humans can quickly read and understand what a hex number represents compared to a binary.